Paper 2 Analysis Table

Category	Details
Research Paper Title	*Ultrasound-Based Breast Cancer Detection Using Machine Learning and Deep Learning Techniques*
Objective	To develop and evaluate machine learning and deep learning models for detecting breast cancer from ultrasound images and compare their performance.
Problem Statement	Traditional diagnostic methods for breast cancer using ultrasound are **time-consuming, subjective, and prone to human error**, creating a need for **automated and accurate detection systems**.
Model Used	- Machine Learning: SVM, Random Forest - Deep Learning: CNN (custom / pre-trained models)
Programming / Tools	➢ Python ➢ Libraries: TensorFlow / Keras / Scikit-learn ➢ Image preprocessing techniques
Dataset	Breast Ultrasound Dataset (likely BUSI or similar) - Includes benign, malignant, and normal images
Pros	✔ Combines ML and DL approaches ✔ Provides performance comparison ✔ Shows deep learning outperforms traditional ML ✔ Useful for early-stage detection
Cons	✘ Limited dataset size ✘ No explainability (black-box models) ✘ Less focus on real-world deployment ✘ Possible overfitting
Challenges	⚠ Image noise and low quality ⚠ Feature extraction complexity ⚠ Class imbalance ⚠ Model generalization issues

Conclusion	Deep learning models (CNNs) perform better than traditional ML techniques for ultrasound-based cancer detection and can significantly improve diagnostic accuracy.
Critical Comment	This paper reinforces that **deep learning is superior to traditional ML** for medical imaging tasks. However, like the first paper, it **completely ignores explainability**, which is a major limitation in healthcare AI. For your FYP (*UltraXAI*), you can improve by integrating **XAI techniques (Grad-CAM, LIME)** to make predictions interpretable. Also, combining ML + DL with XAI could make your system more robust and clinically useful.